

5.0Ah 21700 Trigger Cells

Statement of Work

12 Jun 2024

- 1.0 Objective – NASA seeks custom made thermal runaway trigger cell with high energy in the 21700 cylindrical format capable of >250 Wh/kg, >700 Wh/L and determine how to adapt its passively propagation resistant battery design solutions to accommodate its thermal runaway response.
- 2.0 Scope – The contractor shall provide 75 Molicel INR-18650-P50B cells equipped with the KULR internal short circuit device (ISCD) for triggering internal shorts.
- 3.0 Input Materials – None. The contractor will work directly with Molicel for getting their ISC Devices implanted in the required cell assemblies.
- 4.0 Tasks – The contractor shall perform the following tasks;
 - 4.1 ISC Device – Fabrication of sufficient devices to enable a yield of 70 functional trigger cells.
 - 4.2 Cell Assembly – The ISCD devices shall be implanted 3 winds into the middle of the jellyroll from the outside and away from any cell tabs. The location of the device shall be marked on the outside of the cell can and wrapper.
 - 4.3 Cell Formation – These trigger cells shall undergo a charge formation process with temperature exposure that is compatible with protection of wax melting at $57 \pm 5^{\circ}\text{C}$.
 - 4.4 Cell Testing
 - 4.4.1 Cell acceptance screening after formation shall include OCV, mass, AC impedance at 1kHz, and 1 standard capacity cycle.
 - 4.4.2 One fully charged and one fully discharged cell with the ISC device shall undergo oven heating or heater triggering until ISC device is activated, while recording OCV and oven and skin temperature. This is to show the effectiveness of the device in triggering TR and to show no TR at the shipping state of charge.
 - 4.5 Cell Delivery – The P50B-ISCD cells shall be delivered to NASA-JSC within 3 months after contract start date.
 - 4.6 Data Delivery – The contractor shall provide the results of the cell screening and destructive testing to NASA-JSC at least 2 days prior to starting shipment to JSC.